

Written examination paper for IOS3100 – IoT/OT security
Kristiania University College
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Exam format: Individual written home examination

Final report format: recommended LaTeX or Word format and font 12 with 1.0 spacing. The limit is 15 pages max. The references, table of contents, title page, abstract comes in addition to that.

Grading scale: The Norwegian grading system uses the graded scale A - F, where A is the best grade, E is the lowest pass grade, and F is fail

Weighting: 100% of the overall grade

Support materials: All support materials are allowed

Plagiarism control: We expect your own independent work. Please, use citations and quotations if there is material you want to include in the report together with specific references to the source.

Tasks format: the exam includes three tasks that compose 100% of the overall grade.

Learning outcomes as per course description:

Knowledge. The student...

- can define and explain security issues in IoT systems
- can explain and problematize how the spread of IoT systems in both our daily lives and critical infrastructure will affect security from a societal perspective
- can define and explain selected architectures and protocols in use in IoT devices
- can define and explain security issues in OT systems
- can explain patch problems related to running OT systems, understand what risks may be associated with this from an operational perspective and from a security perspective
- can define and explain selected architectures in use in OT environments
- can explain the challenges and benefits of performing penetration tests in an OT environment

Skills. The student...

- can use tools for static security analysis and code auditing
- can understand the principles behind reverse engineering of hardware and software in an embedded / IoT device, and what effects this has on security
- can carry out a vulnerability analysis / penetration test of IoT devices
- can use tools to carry out vulnerability analyzes and risk analyzes of an OT environment / system
- can analyze protocols used in IoT systems, and use this to uncover security vulnerabilities

General competence. The student...

- can understand and problematise the advantages and disadvantages associated with the emergence of IoT devices and OT systems that are connected to the Internet
- can carry out a security analysis of an IoT product, and understand what threats this product can introduce if it is used in the contexts; private, company and public
- can carry out safety management and risk management of OT projects

Exam Task

Task A – theoretical task (30% of the grade)

A.1: Present two selected practical tasks/projects and corresponding used microcontrollers or microcomputers that you have practised within the course during the exercises. Describe used boards, technologies and the biggest challenges you had and also how you overcame them upon completion of the task.

A.2: Select any two guest lectures we had in this course and reflect on the content of the guest lectures and what is the importance of IoT security that each particular guest lecturer highlighted. Present how this knowledge helped you to better understand concepts from the course.

Task B – use cases (30% of the grade)

B.1: In Lecture 2 (slide number 114) you are given a diagram of “Arduino Temp Sensor” application. Please, extensively describe possible attacks and sketch the attack tree. Use STRIDE threat model to analyze this application and present any two attack scenarios and analyze those using DREAD risk-based framework. The models and relevant approach can be found in Chapters 2/3 of the main textbook “*Practical IoT Hacking: The Definitive Guide to Attacking the Internet of Things*” by Fotios Chantzis et al.

B.2: During the main lectures, there were multiple demonstrations of various software and hardware. Select any two demos and describe the security concerns of each selected technology, and how it can be attacked, protected, or whether the technology you selected can be used to protect the IoT devices and infrastructure.

Task C – practical task (40% of the grade)

C.1: In Chapter 4 of the main textbook you have learnt about the importance of network assessment in IoT applications. Design and implement a simple MQTT-based IoT application that will include three components: a) a weather monitoring station, b) a server-based cloud solution that aggregates all information and c) a user-based mobile client that demonstrates historical weather conditions. This can be an emulated IoT ecosystem (without physical sensors) using at least two virtual machines OR at least two docker instances. The system should include at least a database, IoT application protocols, a central broker, and a client application. You can use either of the technologies: Mosquitto, MQTT Explorer, MySQL, OpenStack, Python, PHP or others with default configurations. Demonstrate vulnerabilities and possible weaknesses of the system, and perform any types of attacks that you have learnt in the program reflecting the OWASP Top 10 Web Application Security Risks (<https://owasp.org/www-project-top-ten/>). Suggest possible protection.

C.2: Pick any of the projects implemented for any Arduino boards (from <https://github.com/>, <https://projecthub.arduino.cc/>, etc). Download main .ino file from a project and present a simple diagram of functionality implemented in setup() and loop() procedures. Analyze the source code and design and find any potential security weaknesses, vulnerabilities, or attack vectors according to OWASP Top 10 IoT framework ([https://wiki.owasp.org/index.php/OWASP Internet of Things Project](https://wiki.owasp.org/index.php/OWASP_Internet_of_Things_Project)). Present and explain at least two cases of how you are going to harden the security of the selected Arduino project, what kind of mechanisms you will implement and what needs to be changed in the source code or whole setup to make the selected project more secure.